Armadale Kalamunda Group

Podiatry Assessment and Management Guideline

1. Purpose

The purpose of this document is to outline the key components of Podiatric Assessment and Management of Diabetes related foot disease at Armadale Health Service. This will ensure alignment with best practice such as International and National Diabetic foot management guidelines.

2. Applicability

Armadale Health Service (AHS) Podiatry department offers podiatric assessment, diagnosis, management and treatment to eligible adult inpatients and outpatients specifically for those at high-risk foot complications.

High risk foot complications arise from a background of comorbidities including but not limited to; Diabetes Mellitus, Peripheral Neuropathy, Peripheral Arterial Disease (PAD), Chronic Renal Disease and Rheumatoid Arthritis. High risk foot complications are associated with increased risk of hospitalisation, amputation, morbidity, and disability. As a result of these complications, prompt and evidence-based assessment and management are essential for achieving optimal foot ulcer healing and reducing amputations in people with high-risk feet.

This service is eligible only to those patients that fall within the Armadale Health Service catchment area.

3. Guidelines

3.1 New patient history

At AHS every new patient will undergo thorough history taking including:

- Review referral
- Prior medical imaging: Call imaging centre and ask for transfer of images to the public PACs system
- Laboratory results: Contact the GP practice or ask consent from patient for My Health Record access
- 3-point identification check
- Medical History
- Medication History
- Surgical History
- Allergies
- Social History: including drinking/smoking status, support network, type of work, mobility, transport, services in place
- History of presenting complaint

3.2 Neurovascular Assessment

A neurovascular assessment is conducted at the patient's initial appointment and then on a regular basis as indicated by the findings. It will assess the Vascular supply to the feet, the sensation of the feet (Neurological assessment), any presence of foot deformity and a history of current or prior ulceration/amputation. This assessment can be recorded on a Neurovascular Assessment Form (Figure 1) or in written documentation.

A thorough vascular assessment of the feet by an AHS podiatrist will include:

- Palpation of pedal pulses +/- popliteal pulses
- If pedal pulses non-palpable a handheld doppler can be used to detect and grade pedal arteries.
- Observation for ulceration and tissue loss
- Inspection of skin colour and integrity (e.g. haemosiderin, venous changes)
- Assessment of skin temperature (e.g. cool or warm)
- Inspection for presence of oedema
- Assessment of capillary refill time (normal <3 seconds).
- Toe systolic pressures
- If toe pressures cannot be performed an ankle brachial index may be used. Note that results from this assessment can be unreliable due to the presence of arterial calcification
- Questioning about rest pain and claudication

In patients with non-palpable pulses, an ankle brachial index <0.5 and/or toe pressure <30mmHg, Vascular Surgeon referral is indicated.

A thorough neurological assessment of the feet will determine the presence or peripheral neuropathy and will be assessed by:

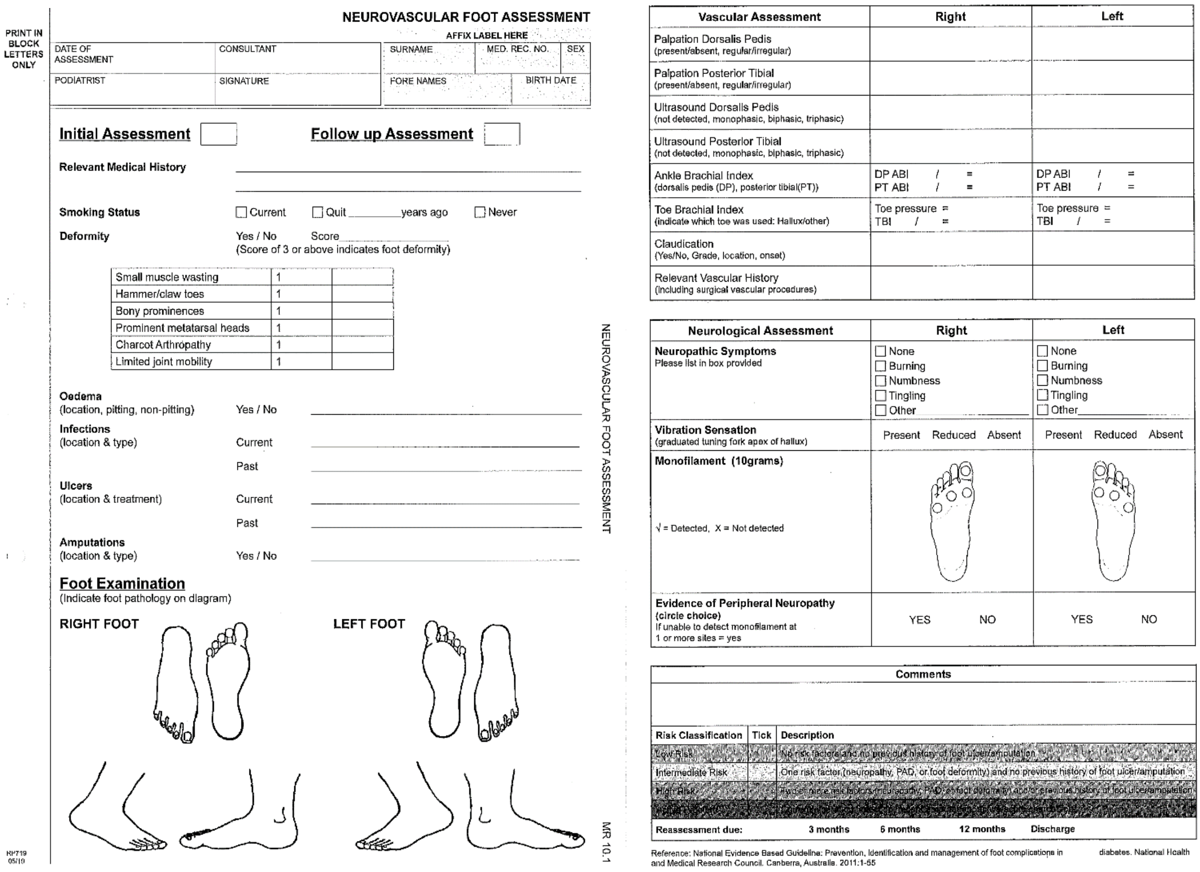
- **10g Monofilament**: Three sites are tested on both feet, hallux, plantar 1st metatarsal head, and plantar 5th metatarsal head. Failure to detect the monofilament at 2/3 sites indicates loss of protective sensation (LOPS).
- **128Hz Tuning Fork**: The tuning fork is applied to 3 bony sites of the foot, dorsal aspect of the 1st distal phalanx, malleolus, and tibial tuberosity. Failure to detect the tuning fork at 2/3 sites indicates loss of vibratory sensation.
- **Ipswich Light Touch Test**: Used when a monofilament or tuning fork is not available, the examiner lightly touches the top of their finger to the plantar foot sites, inability to sense at >2 sites indicate the loss of protective sensation.

Foot Deformity will be assessed by looking for:

- Limited joint mobility
- Presence of Charcot Arthropathy
- Prominent metatarsal heads
- Bony Prominences
- Hammer/Claw toes

- Small Muscle wasting

Figure 1. AHS Neurovascular Foot Assessment form.


NEUROVASCULAR FOOT ASSESSMENT

PRINT IN BLOCK LETTERS ONLY

AFFIX LABEL HERE

DATE OF ASSESSMENT	CONSULTANT	SURNAME	MED. REC. NO.	SEX
PODIATRIST	SIGNATURE	FORE NAMES	BIRTH DATE	

Initial Assessment ☐ Follow up Assessment ☐

Relevant Medical History

Smoking Status ☐ Current ☐ Quit ____ years ago ☐ Never

Deformity Yes / No Score ____ (Score of 3 or above indicates foot deformity)

Small muscle wasting	1		
Hammer/claw toes	1		
Bony prominences	1		
Prominent metatarsal heads	1		
Charcot Arthropathy	1		
Limited joint mobility	1		

Oedema (location, pitting, non-pitting) Yes / No

Infections (location & type) Current / Past

Ulcers (location & treatment) Current / Past

Amputations (location & type) Yes / No

Foot Examination (Indicate foot pathology on diagram)

RIGHT FOOT LEFT FOOT

NEUROVASCULAR FOOT ASSESSMENT

MR 10.1

Vascular Assessment	Right	Left
Palpation Dorsalis Pedis (present/absent, regular/irregular)		
Palpation Posterior Tibial (present/absent, regular/irregular)		
Ultrasound Dorsalis Pedis (not detected, monophasic, biphasic, triphasic)		
Ultrasound Posterior Tibial (not detected, monophasic, biphasic, triphasic)		
Ankle Brachial Index (dorsalis pedis (DP), posterior tibial(PT))	DP ABI / = PT ABI / =	DP ABI / = PT ABI / =
Toe Brachial Index (indicate which toe was used: Hallux/other)	Toe pressure = TBI / =	Toe pressure = TBI / =
Claudication (Yes/No, Grade, location, onset)		
Relevant Vascular History (including surgical vascular procedures)		

Neurological Assessment	Right	Left
Neuropathic Symptoms Please list in box provided	☐ None ☐ Burning ☐ Numbness ☐ Tingling ☐ Other	☐ None ☐ Burning ☐ Numbness ☐ Tingling ☐ Other
Vibration Sensation (graduated tuning fork apex of hallux)	Present Reduced Absent	Present Reduced Absent
Monofilament (10grams) √ = Detected, X = Not detected		
Evidence of Peripheral Neuropathy (circle choice) If unable to detect monofilament at 1 or more sites = yes	YES NO	YES NO

Comments

Risk Classification	Tick	Description
Low Risk		No risk factors and no previous history of foot ulcer/amputation
Intermediate Risk		One risk factor (neuropathy, PAD, or foot deformity) and no previous history of foot ulcer/amputation
High Risk		Two or more risk factors (neuropathy, PAD, and/or foot deformity) and/or previous history of foot ulcer/amputation

Reassessment due: 3 months 6 months 12 months Discharge

Reference: National Evidence Based Guideline: Prevention, Identification and management of foot complications in diabetes. National Health and Medical Research Council. Canberra, Australia. 2011:1-55

This assessment will deem the patient at low, moderate, or high risk of foot disease (Table 1).

Table 1. The IWGDF 2023 Risk Stratification System and corresponding foot screening frequency.

Category	Ulcer risk	Characteristics	Frequency*
0	Very low	No LOPS and no signs of PAD	Once a year
1	Low	LOPS or PAD	Once every 6-12 months
2	Moderate	LOPS + PAD, or LOPS + foot deformity or PAD + foot deformity	Once every 3-6 months
3	High	LOPS or PAD, and one or more of the following: - history of a foot ulcer - a lower-extremity amputation (minor or major) - end-stage renal disease	Once every 1-3 months

Note: LOPS = Loss of Protective Sensation; PAD = Peripheral Artery Disease; * Screening frequency is based on expert opinion, since there is no published evidence to support these intervals

3.3 Foot Ulcer Assessment and Classification

Each clinician will assess the following factors which will guide further evaluation and therapy.

- Type (e.g. neuropathic, neuro-ischamic, ischaemic)
- Cause: Trauma, Pressure, Footwear, Idiopathic
- Site: The anatomical location of the ulcer.
- Size: Measurements length width and depth are taken at each review to track wound progression. This includes probing to bone as indicated.
- Wound base: Sloughy, granulating, necrotic
- Exudate: Considering the type and amount (e.g. serous, haemoserous, purulent)
- Periwound: Callus, fragile, blistering
- Signs of infection: erythema, swelling, pain, malodour

SINBAD Classification: A simple tool recommended for use to facilitate triage and communication of ulcer characteristics between health professionals. See Figure 2.

Figure 2. SINBAD classification system.

Category	Definition	Score
Site	Forefoot Midfoot and hindfoot	**0** **1**
Ischaemia	Pedal blood flow intact: at least one palpable pulse Clinical evidence of reduced pedal flow	**0** **1**
Neuropathy	Protective sensation intact Protective sensation lost	**0** **1**
Bacterial infection	None Present	**0** **1**
Area	Ulcer <1 cm^2 Ulcer ≥1 cm^2	**0** **1**
Depth	Ulcer confined to skin and subcutaneous tissue Ulcer reaching muscle, tendon or deeper	**0** **1**
Total possible score		**6**

Adapted from Ince P et al. Use of the SINBAD classification system and score in comparing outcome of foot ulcer management on three continents. Diabetes Care. 2008;31(5):964-7.

WIfI Classification: The WIFi classification system is used by health professionals with a certain level of expertise and should be updated with a change in any of the criteria. It uses a combination of scores for wound (based on depth of ulcer or extent of gangrene), ischaemia (based on ankle pressure, toe pressure or TcPO2) and foot infection (based on IWGDF/IDSA criteria) to provide a one-year risk for major amputation and one-year benefit for revascularisation, stratified as very low, low, moderate, or high. See Figure 3.

Figure 3. WIFi Classification

Wound

Grade	Ulcer	Gangrene
0	No ulcer **Clinical description:** ischaemic rest pain (requires typical symptoms + ischaemia grade 3); no wound	No gangrene
1	Small, shallow ulcer(s) on distal leg or foot; no exposed bone, unless limited to distal phalanx **Clinical description:** minor tissue loss. Salvageable with simple digital amputation (1 or 2 digits) or skin coverage	No gangrene
2	Deeper ulcer with exposed bone, joint or tendon; generally not involving the heel; shallow heel ulcer, without calcaneal involvement **Clinical description:** major tissue loss salvageable with multiple (≥3 digital amputations or standard TMA ± skin coverage	Gangrenous changes limited to digits
3	Extensive, deep ulcer involving forefoot and/or midfoot; deep, full thickness heel ulcer ± calcaneal involvement **Clinical description:** extensive tissue loss salvageable only with complex foot reconstruction or nontraditional TMA (Chopart or Lisfranc); flap coverage or complex wound management for large soft tissue defect	Extensive gangrene involving forefoot and/or midfoot; full thickness heel necrosis ± calcaneal involvement

Ischaemia

Grade	Ankle-brachial index	Ankle Systolic Pressure (mmHg)	Toe pressure, transcutaneous oxygen pressure (mmHg)
0	≥ 0.80	>100 mmHg	≥ 60 mmHg
1	0.60 - 0.79	70 - 100 mmHg	40 - 59 mmHg
2	0.4 - 0.59	50 - 70 mmHg	30 - 39 mmHg
3	≤ 0.39	< 50 mmHg	< 30 mmHg

Foot infection

Grade	Clinical manifestation of infection	IDSA/PEDIS infection severity
0	No symptoms or signs of infection	Uninfected
1	Infection present, as defined by the presence of at least two of the following items: • Local swelling or induration • Erythema > 0.5 to ≤ 2 cm around the ulcer • Local tenderness or pain • Local warmth • Purulent discharge (thick, opaque to white, or sanguineous secretion) Local infection involving only the skin and the subcutaneous tissue (without systemic signs). Exclude other causes of an inflammatory response of the skin (e.g. trauma, gout, acute Charcot Neuro-osteoarthropathy, fracture, thrombosis, venous stasis)	Mild
2	Local infection (as described above) with erythema > 2cm, or involving structures deeper than skin and subcutaneous tissues (e.g. abscess, osteomyelitis, septic arthritis, fasciitis) No systemic inflammatory response signs (as described below)	Moderate
3	Local infection (as described above) with the signs of SIRS as manifested by two or more of the following: • Temperature > 38°C or < 36°C • Heart rate > 90 beats/min • Respiratory rate > 20 breaths/min or $PaCO_2$ < 32mmHg • White blood cell count > 12,000 or < 4000cu/mm or 10% immature (band) forms	Severe[a]

$PACO_2$: Partial pressure of arterial carbon dioxide, SIRS: systemic inflammatory response syndrome

[a]Ischaemia may complicate and increase the severity of any infection. Systemic infection may sometimes manifest with other clinical findings, such as hypotension, confusion, vomiting, or evidence of metabolic disturbances, such as acidosis, severe hyperglycaemia, new-onset azotaemia.

Adapted from Mills, J.L et al. The Society for Vascular Surgery Lower Extremity Threatened Limb Classification System: Risk stratification based on Wound, Ischaemia and foot Infection (WIfI). Journal of Vascular Surgery, Jan 2014 and Lew, EJ et al. Clinical Application of the Society for Vascular Surgery (SVS) Lower Extremity Threatened Limb Classification System: Risk stratification based on Wound, Ischaemia and foot Infection (WIfI). Wound practice and research, Nov 2014

3.4 Investigations

If there are signs of clinical infection, a wound swab is taken for Microscopy, Culture and Sensitivities.

A baseline x-ray is indicated for a new patient and then as indicated based on wound progression. At AHS there is a PRC clinic available for walk-in x-rays and ultrasound bookings are possible. If further imaging is indicated referral to a Multidisciplinary Foot Ulcer Clinic should be considered.

3.5 Treatment room preparation and proceedings

Prior to each outpatient appointment:

- All surfaces to be cleaned with Clinell wipes
- Clean and stock workspace

During each outpatient appointment:

- Ensure to use Aseptic Non-Touch Technique (ANTT).
- Bare below the elbow
- Gowns for treating patient
- Bluey beneath patient feet and on clinician lap.
- Gloves to be used when providing treatment.
- One blade per wound.
- Sharps containers for disposal of single use sharps and Qliksmarts for disposal/removal of scalpel blades
- Place instruments for sterilisation in the plastic container. Ensure scalpel blades removed.

After each outpatient appointment:

- All surfaces to be cleaned with Clinell wipes
- Broom to sweep debris
- Complete all medical records on Bossnet DMR
- Complete AHS

Treating inpatients:

- Take inpatient trolley
- Perform a risk assessment of the area e.g., patient bed height and patient position. Refer to Work Health and Safety Hub
- Ensure patient privacy
- Single use instruments disposed of on the ward or place used instruments in kidney dish for transport back to podiatry room

Handover Requirements (ISOBAR): The ISOBAR tool is a step-by-step process that provides a sequential approach to giving and receiving handover. Handover involves complex communication of information in relation to patient care, to ensure the optimal delivery of safe and high-quality health care. Effective clinical handover is essential to facilitate safe continuity of patient care and to reduce adverse events. ISOBAR must be used for all verbal and written handover.

- **Identify**: Introduce yourself and your patient
- **Situation**: Briefly explain the problem and reason for the handover
- **Observations**: Your assessment
- **Background**: Brief and concise history for an overall picture of the patient and family
- **Agree a Plan**: The outcome and tasks that need to be completed
- **Read Back**: Given the person receiving the handover the opportunity to ask further questions. Reconfirm any action that will take place.

A letter to the GP is indicated on first review of a patient and on discharge of the patient from AHS Podiatry.

3.6 Debridement

Sharps debridement should be conducted of non-viable tissue (such as peri wound callus, macerated tissue, slough overlying wound bed), to promote wound healing. Sharp debridement of an ulcer is contraindicated where:

- Chronic limb threatening ischaemia is present (absolute toe pressures<30mmHg)
- Pain or not tolerated by the patient
- Dry and stable gangrene or eschar

Low frequency ultrasonic debridement or surgical debridement can be considered on referral to an appropriate Multidisciplinary Foot Ulcer team.

3.7 Dressings

The IWGDF recommends selecting dressings to control exudate and maintain a moist wound environment, as well as considering comfort and cost. All foot ulcers must be continually monitored as their characteristics and dressing requirements change over time.

Considerations:

- Negative Pressure Wound Therapy (NPWT): Can be useful in the management of surgical sites.
- Hyperbaric Oxygen Therapy (HbOT): Can be considered in non-healing ischaemic wounds despite optimal management. Referral and assessment to the Multidisciplinary Foot Ulcer (MDFU)Clinic for assessment first would be indicated.
- Skin graft: Should consideration of a skin graft be indicated, referral to the MDFU Clinic and/or Vascular team(s) is required.

Product Group	Examples	Indications	Uses
Sucrose-octasulfate impregnated dressing	UrgoStart	Non-infected ulcers Neuroischaemic ulcers	Primary dressing May be left on for up to 7 days Must use for 6 weeks to see benefit

Product Group	Examples	Indications	Uses
Hydrogels	Intrasite Intrasite Conformable	Autolytic debridement Wound hydration	Primary dressing
Alginates	Biatain Alginate Kalkostat	Bleeding wounds	Primary dressing
Hydrofibre	Aquacel	Moderate to highly exudating wounds Wicks upwards preventing periwound maceration	Primary dressing
Cadexomer Iodine	Iodosorb paste Iodosorb powder	Infection De-sloughing wounds	Primary dressing
Povidone-Iodine	Inadine Povidone-Iodine Wipes	Infection To keep necrotic wounds/gangrene dry	Primary dressing Not appropriate for large open wounds
Silver Dressings	Aquacel Ag Biatain Ag Acticoat Acticoat Flex	Infection	Primary dressing Activated with water Some products can stain skin Do not use if patient going for imaging. Can be used for a maximum of 3 months at a time
Hypertonic salt	Mesalt	Hypergranulating wounds De-sloughing wounds	Primary dressing Some patients report pain.
Foams	Mepilex Biatain	Moderate exudate Fragile skin	Primary or secondary dressing Can cut to size
Adhesive foams	Mepilex boarder	Shower proof	Primary and secondary dressing Can be worn for up to 7 days

Product Group	Examples	Indications	Uses
Non-adherent	Zetuvit Zetuvit plus	Highly exudating wounds	Primary and secondary dressing Can be used for up to 7 days

3.8 Offloading

The purpose of offloading is to reduce high plantar tissue stress and is a vital aspect of managing foot ulcers. Patient factors must be considered in the application of offloading. At AHS pressure offloading options for plantar foot ulcers should be considered and offered in the following order per IWGDF guidelines:

- A non-removable knee-high offloading device (TCC or iTCC). Contraindications and precautions for this include; falls risk, fluctuant swelling, ischaemia, infection, high levels of exudate. Referral to a tertiary service for application of a TCC is indicated for application of a cast once the patient is screened.
- A removable knee-high offloading device (such as CAM, VACOPED, Air walker)
- An ankle high offloading device (darco shoe)
- Appropriate and well-fitting footwear +/- orthotics. While generally reserved for long term offloading, can be considered if above options unsuitable based on patient factors.
- Felt: generally used in conjunction with other offloading methods to optimise the offloading effect. It should generally not be considered as the primary offloading option except where patient compliance with other devices is questionable.

Management of a heel ulcer or heel pressure injury should also incorporate offloading at night or whilst in bed. Commonly used heel offloading to consider if a pillow under the ankles, air mattress, PRAFO or a heel-lift suspension boot. Selection of appropriate heel offloading devices will be dependent on many clinical factors and is at the discretion of the clinician, including tolerance, skin integrity, falls risk, whether the patient may require walking to bathroom during the night, ability to don and doff, exudate levels and location/aetiology of the ulcer. These will be ordered on an as needed basis.

Charcot Restraint Orthotic Walker (CROW) This device could be considered as effective as a total contact cast if made irremovable. The CROW is a custom-made device and as such is costly. CROW’s should be considered as a medium to long term option for patients with sub-acute or chronic Charcot Neuroarthropathy with associate significant foot deformity or where severe foot deformity prevents use of other offloading devices. Referral to an appropriate service for manufacture is indicated at AHS.

3.9 Escalation

Should there be concerns with a patient that requires medical assessment the options are as follows:

Treatment of PAD:

Referral either direct to Vascular service or to the Multidisciplinary Foot Ulcer Clinic may be indicated if suspected diminished arterial supply in the context of having a foot ulcer.

Consider asking the GP for a doppler ultrasound or CTA prior to referral. Ensure toe pressures and pedal pulse assessment has been completed prior to referral.

Treatment of Limb Threatening Infection or Ischemia:

Urgent phone call to Vascular team and direct patient per advice of the Vascular team. If unable to reach for any reason, direct patient to closest Emergency Department with handover and continue to contact Vascular team.

Treatment of infection:

Should an ulcer present with local infection, consider wound swab to guide antimicrobial therapy. In the absence of an endorsed prescriber, a letter with the supporting results of the wound swab should be sent to the GP for consideration of oral antibiotics.

If the patient is exhibiting systemic signs of infection (e.g. fevers, chills, nausea, vomiting), the patient must be escalated to the closest Emergency Department.

Referral to the MDFU clinic may be indicated in the presence of underlying osteomyelitis.

Chronic Foot ulcer:

Should a foot ulcer be present with nil improvement despite >6 weeks of standard therapy, consider referral to the MDFUC which allows for access to team expertise.

4. Roles and responsibilities

It is the role of every Podiatrist at AHS to appropriately assess, manage and escalate patients referred to the Podiatry clinic. Should you have any concerns please raise to a Senior Podiatrist.

5. Legislative context

The following documents are required to give effect to this policy:

- Work Health Safety Act

6. Supporting information

The following documents support the implementation of this policy:

- AKG Wound Management Procedure
- AKG Specimen Collection and Pathology Results Procedure

7. Compliance, monitoring and evaluation

While this is a guideline, it will be monitored sporadically with clinical documentation audits.

8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 (Comprehensive Care) – Action 5.10: Clinicians use relevant screening processes:
- Standard 5 (Comprehensive Care) – Action 5.11: Clinicians comprehensively assess the conditions and risks identified through the screening process
- Standard 6 (Communicating for Safety) – Action 6.04: The health service organisation has clinical communications processes to support effective communication when:

- Standard 8 (Recognising and Responding to Acute Deterioration) – Action 8.06: The health service organisation has protocols that specify criteria for escalating care, including:

9. References

1. Diabetes Feet Australia. (2021). 2021 Australian Guideline on prevention of Foot Ulceration (Version 1.0 140921). New Guidelines - Diabetes Feet Australia
2. International Working Group on the Diabetic Foot. (2023). IWGDF 2023 Guidelines. All IWGDF Guidelines (2023 update) - IWGDF Guidelines

10. Glossary

The following definitions are relevant to this policy.

Term	Definition
High risk	A person with Loss of Protective Sensation or Peripheral Arterial Disease and one or more of the following: • History of a foot ulcer • A lower-extremity amputation (minor or major) End-stage renal disease
Moderate risk	A person with: • Loss of Protective Sensation + Peripheral Arterial Disease • Loss of Protective Sensation + Foot Deformity Peripheral Arterial Disease + Foot Deformity
Low risk	A person with: Loss of protective sensation or Peripheral Arterial Disease
Very-Low risk	A person without loss of protective sensation or Peripheral Arterial Disease.

11. Document owner

Enquiries relating to this guideline may be directed to:

Title: Podiatry Coordinator
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12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V2	25/07/2025	25/07/2028	Scheduled review

13. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Danielle Kilmurray – Director Allied Health and Ambulatory Care
Authorisation date	30/06/2025
Published date	25/07/2025